

Zhang, Wei

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Chinese citizen; Singapore Permanent Resident

Education

- 09/2013 - 06/2017 **B.S., Engineering Physics, Tsinghua University**, Beijing, China.
- 09/2017 - 01/2019 Program of Interdisciplinary Courses in *Computer Science*, **University of Chinese Academy of Sciences**, Beijing, China.
- 09/2017 - 06/2022 **Ph.D.** in *Particle Physics and Nuclear Physics (major in Medical Image Analysis)*, **University of Chinese Academy of Sciences**, Beijing, China. **M.S.** received in 2019.

Work Experience

- 2015 Research Assistant at the student training project in **Tsinghua University**, Beijing, China.
- 2016 Internship at **GSAFETY Technology Co., Ltd.**, Beijing, China.
- 2019 Internship at the department of neurology in **Beijing Xuanwu Hospital, Capital Medical University**, Beijing, China.
- 2021 Internship at the department of neurology in **Beijing Tiantan Hospital, Capital Medical University**, Beijing, China.
- 2017 - 2022 Research Assistant at **University of Chinese Academy of Sciences**, Beijing, China.
- 2022 - present Research Fellow at Nanyang Technological University, Singapore

Representative First-Author Papers & Posters

1. **Zhang, W.**, et al., Adult-to-infant unidirectional neural coupling mediates selective social learning in infants from the United Kingdom and Singapore. *Nature Communications*, 2026
2. **Zhang, W.**, et al., Spatiotemporal Encoding Signatures of Covert Speech Revealed by Source-Localized EEG and fMRI. *Communications Biology*, 2026
3. **Zhang, W.**, et al., Revealing the Spatiotemporal Brain Dynamics of Covert Speech compared with Overt Speech: A Simultaneous EEG-fMRI Study. *NeuroImage*, 2024
4. **Zhang, W.**, et al., Investigating Sea-Level Brain Predictors for Acute Mountain Sickness: A Multi-Modal MRI Study before and after High-Altitude Exposure. *American Journal of Neuroradiology*, 2024
5. **Zhang, W.**, et al., Deep Learning With 18F-Fluorodeoxyglucose-PET Gives Valid Diagnoses for the Uncertain Cases in Memory Impairment of Alzheimer's Disease. *Frontiers in Aging Neuroscience*, 2021

6. **Zhang, W.**, et al., Modeling brain functional networks using graph neural networks: A review and clinical application. *ICCK Transactions on Intelligent Systematics*, 2024
 7. **Zhang, W.**, et al., Speaker-Listener Neural Coupling Explains Gating Effect of Gaze on Infant Statistical Learning. *OHBM Annual Meeting*, 2025
 8. **Zhang, W.**, et al., Multi-Site Voxel-Based Morphometry: Reliability Evaluation and a Harmonization Method. *International Conference on Medical Imaging Physics and Engineering (ICMIPE 2019) & The 9th National Annual Meeting of Medical Imaging Physics (NAMMIP)*, 2019
- <https://scholar.google.com/citations?user=Kqc41x4AAAAJ>

Research Interests

Multimodal neuroimaging

AI and brain-computer interfaces

Neurolinguistics

Language and social neuroscience

Reviewing & Teaching Activities

Invited reviewer for *NeuroImage*, *Imaging Neuroscience*, *Developmental Cognitive Neuroscience*, *Scientific Reports*, *Frontiers in Human Neuroscience*, *IEEE Journal of Biomedical and Health Informatics*, and *Signal, Image and Video Processing*

2024 - 2025 Guest lecturer in *HP4021 Laboratory in Human and Animal Neuroscience*, Nanyang Technological University

2026 AI tutorial instructor in *GP8000 AI Literacy*, Graduate College PGR ICC, Nanyang Technological University

Professional Membership & Talks

2024 - 2025 Member of American Society of Neuroradiology (ASNR)

2026 Invited Young Speaker, AP-CCN 2026, Singapore

Awards and Honors

2013 Leading talent scheme, Tsinghua University, Beijing, China.

2018 - 2021 National Scholarship for Postgraduate (¥10,000 direct), Institute of High Energy Physics, Chinese Academy of Sciences

2022 Dean's Scholarship (¥13,665 direct, TOP 1 in each department, 300 out of all postgraduates in CAS), Chinese Academy of Sciences

2022 Zhu Li Yue Hua Excellent Doctoral Student Award (¥5,000 direct, TOP 1 in each department, 300 out of all postgraduate candidates in CAS), Chinese Academy of Sciences